

# STATEMENT OF PURPOSE

**Applicant:** Xuan Thanh Trinh

**Program:** Ph.D. in Electrical, Electronics and Computer Engineering

**Proposed Advisor:** Assistant Professor Van-Hai Bui

**Institution:** University of Michigan-Dearborn

## 1. Introduction

As a final-year Electrical Engineering student at Hanoi University of Science and Technology (HUST), my research at the Power Grid and Renewable Energy (PGRE) Laboratory focuses on developing distributed mathematical optimization models for decentralized energy management. While this foundation drove my graduation thesis on microgrid resilience, I recognized that classical optimization alone cannot scale to the real-time computational demands of modern power systems. To overcome these computational bottlenecks, I require advanced training in scalable, cyber-physical control systems. The Ph.D. program at UM-Dearborn perfectly aligns with this need, providing the rigorous framework I need to become a highly specialized research engineer capable of designing decentralized architectures for high renewable penetrations.

## 2. Academic Background

Ranking top 3 in HUST's Advanced Program in Electrical Engineering and Renewable Energy, I trained to deconstruct complex grid architectures into tractable engineering problems. Across diverse projects, I approached these challenges from multiple technical dimensions, for instance, by developing predictive models to quantify system uncertainties and building basic circuits to understand hardware limitations. Maintaining this multi-faceted perspective directs my independent research trajectory and mitigates critical theoretical and physical blind spots.

Safeguarding legacy power systems against cascading failures requires a paradigm shift from rigid, top-down dispatch architectures to decentralized, autonomous microgrid communities. Recognizing this necessity, my research at the PGRE lab focused on architecting a decentralized electricity market. The core objective was to subdivide adjacent urban districts into autonomous microgrids capable of trading energy with one another to achieve global economic optimization while maintaining practical stability conditions.

A truly equitable energy transition requires decentralized markets that guarantee physical reliability alongside economic fairness. By integrating rigorous grid physics (AC-OPF) with distributed optimization (ADMM), I developed a peer-to-peer trading framework for both radial and ring-connected topologies. Beyond merely reducing overall operational costs compared to independent microgrid operation, this architecture strictly bounded voltage levels within the critical  $[0.95, 1.05]$  pu threshold and accounted for line losses, ensuring that community-level energy trading does not compromise the structural integrity of the wider public grid.

Furthermore, modern grid architectures must transcend static optimization to dynamically shield communities from catastrophic local failures. To achieve this fault tolerance, I layered Model Predictive Control (MPC) over the trading baseline to coordinate emergency power exchanges and backup diesel generation. During simulated localized outages, this temporal coordination empowered neighboring urban districts to physically and economically assist a failing grid. By successfully reducing forced load shedding from 15 MWh down to just 3 MWh, this framework demonstrated how algorithmic resilience directly translates to uninterrupted power for essential civilian infrastructure.

However, the inherent computational latency of non-linear physical constraints remains a critical barrier to deploying these models in real-time. While my mathematical models guaranteed strict physical safety and global convergence, the processing overhead of continuously re-optimizing non-linear AC-OPF equations rendered classical solvers unscalable for second-by-second dispatch. In a world rapidly transitioning to volatile renewable energy sources, this computational bottleneck poses a direct threat to grid stability. Overcoming this limitation through learning-based AI control is not just a technical necessity, but a societal imperative—and it directly motivates my decision to pursue a Ph.D.

### 3. Research Interests

To resolve the trade-off between computational tractability and physical grid security, my Ph.D. research will focus on architecting Physics-Informed AI frameworks for real-time power system dispatch. While pure data-driven models accelerate computational speeds, their deployment frequently fails because unconstrained black-box policies inherently violate non-linear AC power flow equations and thermal limits. I intend to synthesize these paradigms by deploying my deterministic optimization models in a dual capacity: as expert benchmarks to generate mathematically rigorous optimal dispatch datasets, and as differentiable safety filters—leveraging convex relaxations like SOCP—that project AI-generated setpoints into a verifiable feasible region. By incorporating structural grid constraints into both the offline training and online execution agents, I aim to pioneer certifiably safe and physically resilient dispatch architectures.

### 4. Academic and Research Alignment

Realizing that architecting Physics-Informed AI frameworks requires an ecosystem that seamlessly bridges advanced machine learning with rigorous mathematical optimization, I found the Ph.D. program at UM-Dearborn to be the perfect catalyst for my research. To construct mathematically guaranteed safety filters for these AI agents, I require a stringent theoretical foundation. The interdisciplinary scope of the EECE curriculum will systematically bridge my theoretical gaps, providing the comprehensive mathematical and computational toolkit necessary for this architecture through advanced courses such as IMSE 505 (Optimization) and ECE 5001 (Analytic and Computational Math). Furthermore, the robust computational facilities at UM-Dearborn, paired with its strategic ties to regional energy sectors, provide the exact environment I need to rigorously test and validate these theoretical architectures against real-world grid scenarios.

Within this dynamic ecosystem, my research trajectory strongly aligns with the ongoing work of Prof. Van-Hai Bui in autonomous energy markets. His recent publications on FairMarket-RL and XRL-LLM investigate the integration of Large Language Models into multi-agent energy trading, an area highly relevant to my interest in learning-based policies. By joining his laboratory, my foundation in mathematical optimization will allow me to seamlessly support his ongoing research while advancing my own theoretical frameworks. Crucially, Prof. Bui's mentorship and profound expertise in AI surrogates will provide the exact guidance I need to successfully integrate deterministic AC safety filters into learning-based policies, ultimately realizing my vision of a physically resilient smart grid.

### 5. Career Objectives

Following the completion of my Ph.D, I intend to deepen my expertise as an R&D scientist within the U.S. energy sector before ultimately returning to Vietnam to contribute to the technical modernization of our national grid infrastructure. By repatriating advanced learning-based control methodologies, my long-term vision is to help transition our power architectures toward autonomous, resilient networks capable of stabilizing the rapid integration of clean energy sources. Ultimately, the Ph.D. program at the University of Michigan-Dearborn provides the exact interdisciplinary ecosystem and mentorship necessary to launch this career trajectory.